

# WASP-104b

R-2359

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-104\_200113 · created 2026-08-19T23:05:45+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458861.9292 +/- 0.0026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.02 +/- 0.04                   |
| Transit depth (Rp/Rs)^2                           | 0.042 +/- 0.16 [%]              |
| Orbital Inclination (inc)                         | 83.62 +/- 2.94                  |
| Airmass coefficient 1 (a1)                        | 1.117 +/- 0.012                 |
| Airmass coefficient 2 (a2)                        | -0.099 +/- 0.038                |
| Scatter in the residuals of the lightcurve fit is | 1.08 %                          |
| Best Comparison Star                              | #1 - [544, 157]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.054 +/- 0.03                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                         |
|---------------------------|-----------------------------------------|
| Rp/R* fit vs published    | 0.02 ± 0.04 vs 0.1214 (deviation 2.53σ) |
| Duration fit vs published | 0.054 d vs 0.0742 d (deviation 0.67σ)   |
| Mid-time O–C              | 6.8 min                                 |
| Depth SNR                 | 0.5                                     |
| Residual scatter          | 1.08 %                                  |

### Light curve

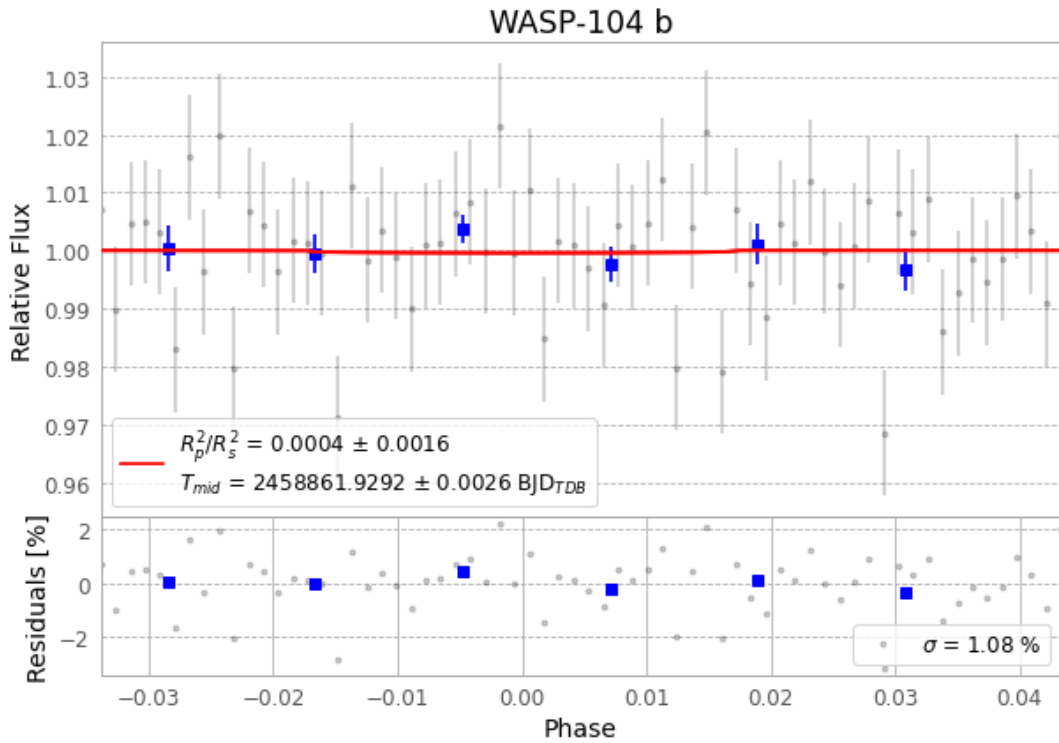


Figure 1 — FinalLightCurve\_WASP-104 b\_2020-01-13.png (SHA-256 3946c01370a398d6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [155, 254], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c608cdd756090677...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

66 FITS frames (43 MB), WASP-104200113084512.FITS → WASP-104200113120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-200113.log (5ef58b25a39d...), FinalLightCurve\_WASP-104 b\_2020-01-13.png (3946c01370a3...), FinalLightCurve\_WASP-104 b\_2020-01-13.csv (8756d26457e0...)

## Compute resources

Wall time 153.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:05Z.